

ASIC Design Flow Tutorial

Lab 1: Logic Simulation. VCS

A. Laboratory tasks

Note: source /home/share/Env/synopsys_setup.env to open all tools.

1. Access into the directory: ~/TH_TKVM_CLC/Digital_Synopsys/Sim/RTL and write 2 files, including a design file (.v) and a testbench file(.v) for the 8-bit Johnson Counter design with the following truth table:

Clock	Cycle	O0	O1	O2	O3	O4	O5	O6	O7
↑	1	0	0	0	0	0	0	0	0
↑	2	1	0	0	0	0	0	0	0
↑	3	1	1	0	0	0	0	0	0
↑	4	1	1	1	0	0	0	0	0
↑	5	1	1	1	1	0	0	0	0
↑	6	1	1	1	1	1	0	0	0
↑	7	1	1	1	1	1	1	0	0
↑	8	1	1	1	1	1	1	1	0
↑	9	1	1	1	1	1	1	1	1
↑	10	0	1	1	1	1	1	1	1
↑	11	0	0	1	1	1	1	1	1
↑	12	0	0	0	1	1	1	1	1
↑	13	0	0	0	0	1	1	1	1
↑	14	0	0	0	0	0	1	1	1
↑	15	0	0	0	0	0	0	1	1
↑	16	0	0	0	0	0	0	0	1

- Example (code for the design) – johnson.v

```
module johnson (clk, r, out);

    parameter size=7;
    input clk;
    input r;
    output [0:size] out;

    reg [0:size] out;

    always @(posedge clk or posedge r)
        begin
            if (r)
                out= 8'b0000_0000;
            else
                out={~out[size],out[0:size-1]};
        end

endmodule
```

- Example (code for testing) – johnson_test.v

```
`timescale 1ns/1ns
module jonson_test;
parameter size=7;
reg clk;
reg r;
wire [0:size] out;
johnson xjohnson(clk,r,out);
//initial xjc.size=size;
initial begin
    clk=1;
    r=0;
    #5 r=1;
    #5 r=0;
    #100 $finish;
end
initial $monitor($time,,"[%b]\t%b\tq=%b",clk,r,out);
always #2 clk=~clk;
endmodule
```

2. Compile johnson.v RTL source and Johnson_test.v testbenches using VCS with the following command:

```
% vcs johnson_test.v johnson.v -gui -debug_all
```

-gui Enables DVE
-debug_all Enables command line debug including line tracing (optional)

If compilation is successful, open DVE from the command line:

```
% dve &
```

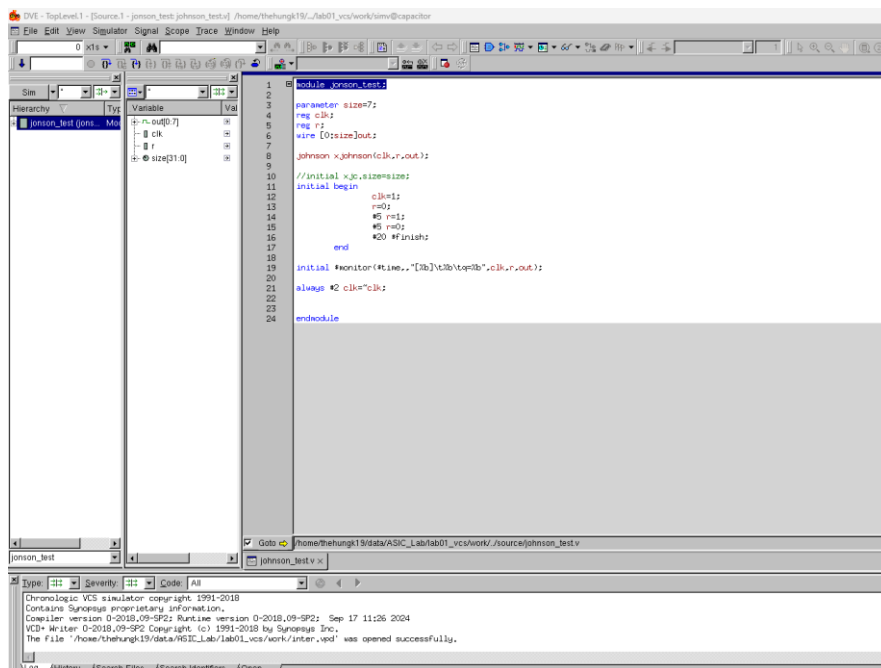


Fig.1. DVE Top-level GUI window

3. Setup DVE
 - From menu bar select *Simulator > Setup*,

- In "Simulator executable" press "Browse..." and select "simv" file in the current directory

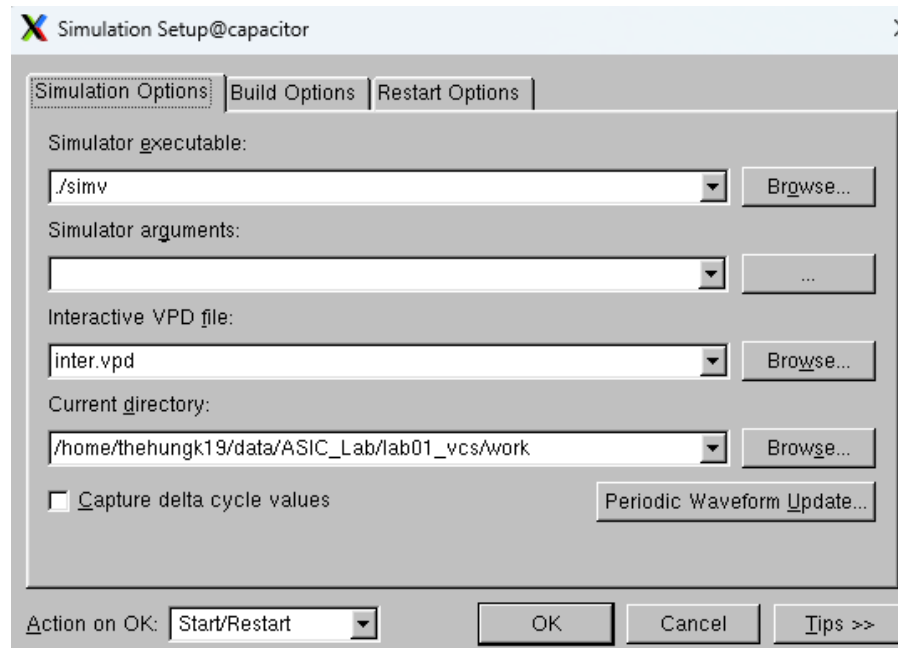


Fig.2. Simulation setup

The Simulation setup has the following options (Fig.2.):

Simulator Executable - specifies the name of a simulator executable.

Simulator arguments - identifies the simulator arguments.

Interactive VPD file - specifies the name of the VPD file. VPD files (design database files) are platform- independent, versioned files into which it is possible to dump the selected signals during simulation. DVE gets hierarchy, value change, and some assertion information from these files.

Current directory - specifies the full path of the simulator executable.

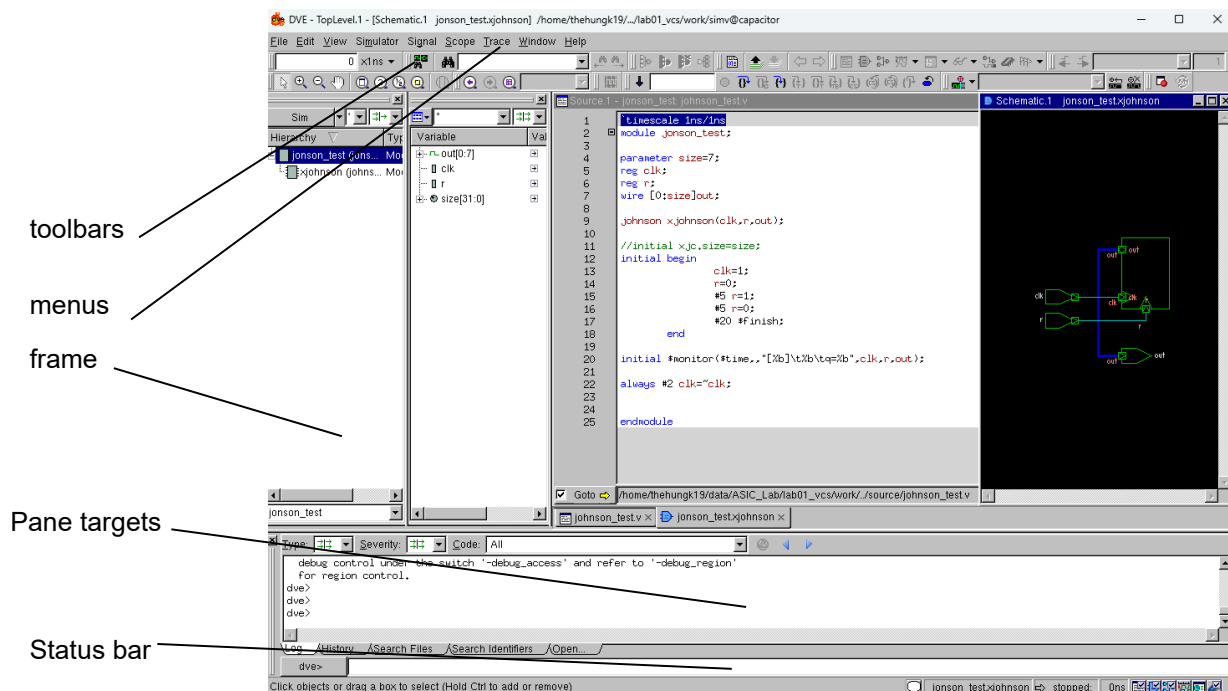


Fig.3. Overview of DVE window

A TopLevel window contains:

1. Frame
2. Menus
3. Toolbars
4. Status bar
5. Pane targets.

Viewing a Waveform

Hierarchy Pane - displays the scope hierarchy of the design.

Data Pane - displays the variables of the selected scopes of the Hierarchy pane.

Source View - Displays the source code and supports source code relative features, such as tracing driver or load, and setting line breakpoints.

List View - provides a table view to display the values of signals over time.

Schematic View - provides a module-based schematic to display the connectivity of the object.

Assertions View - displays the summary of assertion results of simulation including the success, failures, and the incomplete ones.

To view waveform information for signals in the Wave view (Fig.4).

1. Select a scope or object from the Hierarchy pane, Data pane, Source view, List view, Schematic view, or Assertion view.



2. Click the Add to Waves icon in the toolbar.

Using Simulator Menu Commands Runs the simulation until a breakpoint is hit, the simulation finishes, or for the duration specified in the set Continue Time dialog box or toolbar time entries.



When the simulation is running, this icon is activated. Click to stop the simulation.



When the simulation completes, the signals appear automatically. Press the “F” key to expand the full screen.

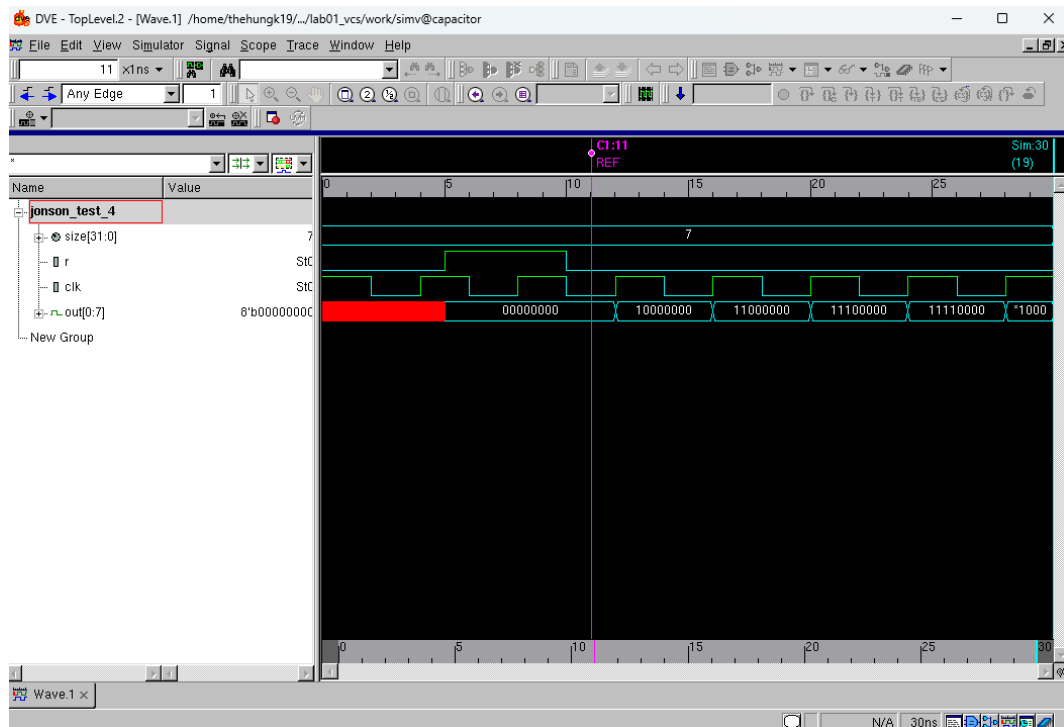


Fig.4. WaveView window

Tracing Drivers and Loads

1. Trace drivers and loads of a signal at any time to see the drivers/loads that caused a value change and see all the drivers/loads that possibly contributed to a signal value.
2. Select a signal in a view or pane. For example, Data pane, Source view, List view, etc.
3. Right-click and select Trace Drivers or Trace Loads. When a driver is traced, a new Driver pane will be created if none exists in the current top level frame. If a driver pane exists, the driver information will be added to the top of the list.

Additionally, the first driver will be highlighted in the Source view and annotated with a blue node in the gutter. In the Wave view, double-click on a waveform to see its drivers. (Fig. 5)

Link the Driver panes to Source view in the same top level frame and Path Schematic view. The Link to radio buttons, at the right top of the pane, shows the current linked windows. By linking a Source and Schematic view, when the object is selected in the Drivers pane, the object will also be selected in the linked views.

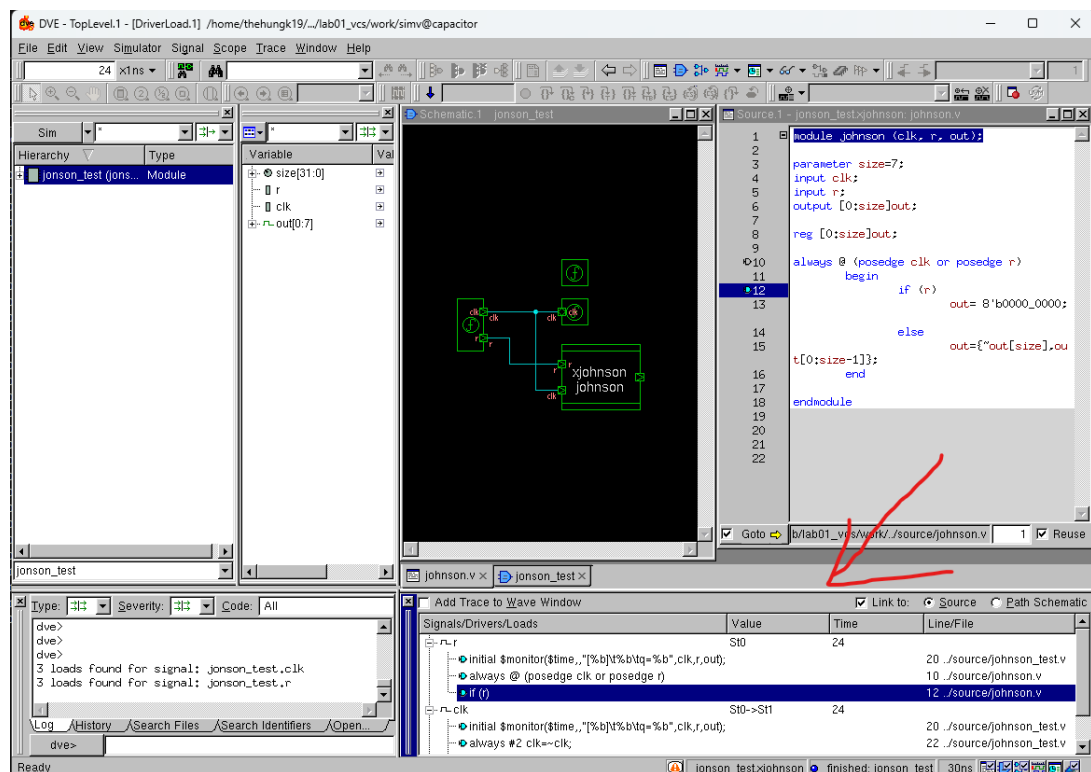


Fig.5. Top-level window with tracing drivers and loads

To exit DVE, write exit in the command line on the console.

```
DVE> exit
```

B. Homework

1. Design the **combinational** circuit and testbench:

$$Y = \overline{A}.\overline{B}.\overline{C} + A.\overline{B}.\overline{C} + A.\overline{B}.C.$$

2. Design the **counter** circuit and testbench with the following description:

Index	Pin name	No. Bit	Direction	Function
1	clear	1	input	
2	clock	1	input	
3	state	8	output	data_out